

CoRoT-2b

R-1584

Benchmark run · workflow W8-benchmark-deep · record deep-bench_CoRoT-2_210610 · created 2026-08-02T00:15:46+00:00

BENCHMARK: MARGINAL

Results

Mid-Transit Time (Tmid)	2459375.8957 +/- 0.0016 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.171 +/- 0.015
Transit depth (Rp/Rs)^2	2.91 +/- 0.5 [%]
Orbital Inclination (inc)	86.92 +/- 1.39
Airmass coefficient 1 (a1)	1.037 +/- 0.018
Airmass coefficient 2 (a2)	-0.027 +/- 0.03
Scatter in the residuals of the lightcurve fit is	1.74 %
Best Comparison Star	#2 - [346, 120]
Optimal Aperture	4.07
Optimal Annulus	13.89
Transit Duration (day)	0.0917 +/- 0.0049

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.171 ± 0.015 vs 0.1667 (deviation 0.27σ)
Duration fit vs published	0.0917 d vs 0.095 d (deviation 0.63σ)
Mid-time O–C	5.9 min
Depth SNR	10.7
Residual scatter	1.74 %

Light curve

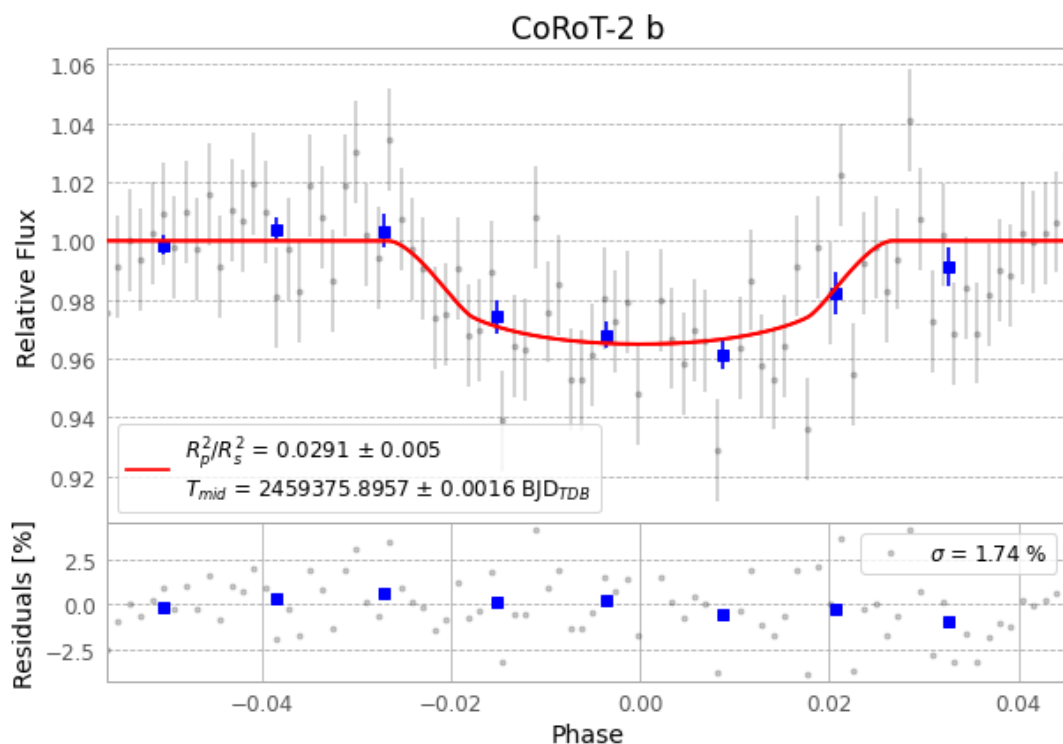


Figure 1 — FinalLightCurve_CoRoT-2 b_2021-06-10.png (SHA-256 eb541bbb32e54313...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [401, 94], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 9d0f1af7a3dfb8df...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

86 FITS frames (56 MB), CoRoT-2210610070012.FITS → CoRoT-2210610111512.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-CoRoT-2-210610.log (e19216550248...), FinalLightCurve_CoRoT-2 b_2021-06-10.png (eb541bbb32e5...), FinalLightCurve_CoRoT-2 b_2021-06-10.csv (8f8c6e5f0934...)

Compute resources

Wall time 145.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-02 00:15Z.